Benjamin Nudelman

646-477-1086 | bnudelman2@gmail.com | linkedin.com/in/nudelman-ben/ | github.com/bnudelman4 | Brooklyn, NY

Education

Cornell University — Ithaca, NY
Bachelor of Science in Computer Science — *Expected May 2027*
Minor in Operations Research and Information Engineering (ORIE), Minor in Applied Mathematics

- GPA: 3.7
- Relevant Coursework: Machine Learning, Algorithms, Linear Algebra, Data Science, Data Structures and Functional Programming, iOS App Development, Object-Oriented Programming and Data Structures, Multivariate Calculus, Computer System Organization and Programming, Differential Equations, Compilers, Operating Systems

Experience

Cornell Custom Silicon Systems (C2S2) — Sep. 2025 – Present
Embedded Software Engineer — *Ithaca, NY*

- Designed and deployed an advanced neural network using TensorFlow Lite on a microcontroller to implement a secure two-factor authentication system using on-chip microphone and IMU sensor data
- Write firmware in C and C++ that runs directly on microcontrollers, integrating software and hardware and developing collaboratively with other subteams
- Use libraries such as TinyML to deploy lightweight neural networks directly on chip

Cornell Cybersecurity Club — Sep. 2024 – Present
Cybersecurity Software Engineer — *Ithaca, NY*

- Created ethical hacking algorithms using C, Python, BASH, and Javascript
- Participated in Capture the Flags and practice hacking challenges both individually and within teams
- Attended workshops and lectures from employees and researchers for MITRE and MIT Lincoln Laboratories

Massachusetts Institute of Technology — Feb. 2023 – Aug. 2023
Student Researcher, Beaver Works Summer Institute — *Cambridge, MA*

- Designed and programmed secure embedded systems for a software installer/updater for a car, in teams of 4 or 5
- Created software for emulated chips and interacted with different hardware and software components of the chip, applying knowledge of C, assembly language, Python, BASH and secure encryption algorithms
- Competed in hackathons against peer groups under time constraints to implement defense strategies and defend against ethical hacking mechanisms, and participated in similar Capture the Flag events

Projects

In21 | *Swift, React Native, Firebase, Docker, Tailwind CSS*, Git — June 2025 – Present

- Co-founded a productivity application catered to teenagers and young adults interested in self-improvement
- Implemented an efficient backend for storing user contact information and relevant metrics from in-app activities and surveys, created a messaging system between users and an email verification / sending system
- Utilizing Figma to design the page flows, then directly implementing the front end and APIs using React Native

UniRides | *Swift, UIKit, REST APIs, Codable, MVC, JSON*, Git — Nov. 2024 – Mar. 2025

- Built an iOS carpooling application enabling university students to discover, join, and create rides between campus and home locations
- Implemented the frontend using UIKit and MVC architecture
- Integrated the frontend with backend REST APIs, defined JSON schema and implemented Codable models with custom CodingKeys to map backend responses

SimplifyCS | *Next.js, Node.js, MongoDB, Prisma, REST APIs, bcrypt*, Git — Feb. 2023 – Jan. 2024

- Created a full-stack CS education platform with a group of 3, developing a backend using Next.js API routes and MongoDB via Prisma, modeling a curriculum hierarchy from courses to units to lessons
- Built responsive frontend interfaces using Next.js and React, implementing dynamic pages for course navigation, lesson viewing, and content creation
- Designed CRUD endpoints and a polymorphic activity system using indexed references to dynamically support multiple lesson types such as articles, quizzes, and assignments
- Implemented secure authentication using bcrypt, token-based sessions, and email verification

Technical Skills

Languages: Java, Javascript, Python, C/C++, HTML/CSS, x86/RISCV Assembly Language, OCaml, Swift
Frameworks/Databases: React, Node.js, Prisma, Firebase, Flask, JUnit, postgreSQL, MongoDB
Developer Tools/Libraries: Git, Docker, VS Code, Visual Studio, IntelliJ, pandas, NumPy, Matplotlib, Linux/Unix